

PROJECT REPORT:

Loan Approval Prediction Using Machine Learning

SUBMITTED BY:

RAMEEN KHAN (54)

BISMA NISAR (6)

AMNA AZEEM(3)

CLASS:

PITP DATA SCIENCE GROUP-1

COURSE INSTRUCTORS:

SYED SHAHZAIN HASSAN

HASNAIN NAQVI

1. Introduction

Loan approval is a crucial decision-making process in financial institutions, where multiple applicant attributes are evaluated to minimize risk. Traditional methods rely heavily on manual judgment, which may lead to inconsistency and inefficiency.

This project applies data analysis and machine learning techniques to identify the key factors influencing loan approval and to build predictive models that assist in decision-making.

2. Objectives

- To perform detailed Exploratory Data Analysis (EDA)
- To identify the most influential variables affecting loan approval
- To preprocess and transform the dataset for better modeling
- To build and compare classification models
- To evaluate model performance using appropriate metrics

3. Dataset Description

The dataset consists of applicant-level information used for loan approval decisions.

Table 1: Variables Description

Variable	Type	Description
Gender	Categorical	Male/Female
Married	Categorical	Marital status
Dependents	Categorical	Number of dependents
Education	Categorical	Graduate/Not Graduate
Self_Employed	Categorical	Employment status
ApplicantIncome	Numerical	Applicant income
CoapplicantIncome	Numerical	Co-applicant income
LoanAmount	Numerical	Loan amount requested
Loan_Amount_Term	Numerical	Loan repayment term
Credit_History	Numerical	0 = Bad, 1 = Good
Property_Area	Categorical	Rural/Semiurban/Urban
Loan_Status	Target	Y = Approved, N = Rejected

4. Data Preprocessing

Raw data contained missing values and inconsistent formats, which were handled as follows:

- Missing values in categorical variables were filled using **mode**
- Missing values in numerical variables were filled using **mean**
- Categorical variables were encoded into numeric format
- Irrelevant column (**Loan_ID**) was removed

5. Feature Engineering

To improve predictive performance:

- **TotalIncome = ApplicantIncome + CoapplicantIncome**
- Log transformation applied to reduce skewness:
- ApplicantIncome_log
- LoanAmount_log
- TotalIncome_log

These transformations help stabilize variance and reduce the effect of extreme values.

6. Exploratory Data Analysis (EDA)

EDA was conducted to understand patterns and relationships in the dataset

Key Insights:

- **Loan Approval Rate:** ~69% approvals → slight class imbalance
- **Credit History:** Strongest predictor; applicants with good credit history are far more likely to be approved
- **Income Distribution:** Highly skewed with extreme outliers
- **Loan Amount:** Right-skewed, improved after log transformation
- **Property Area:** Semiurban applicants show higher approval rates
- **Total Income:** More informative than individual income variables

Table 2: Important Observations

Feature	Observation
Credit_History	Strong positive relation with approval
ApplicantIncome	Contains extreme outliers
LoanAmount	Right-skewed distribution
Property_Area	Semiurban shows higher approvals

Visualizations Used

- Count plots (Loan_Status vs categorical variables)
- Histograms (distribution analysis)
- Boxplots (outlier detection)
- Scatter plots (income vs loan amount)
- Correlation heatmap

7. Model Development

The dataset was split into training (80%) and testing (20%)

.

Models Applied:

Model	Purpose
Logistic Regression	Baseline linear model
Decision Tree	Non-linear pattern detection
Random Forest	Ensemble model for improved accuracy

8. Model Evaluation

Models were evaluated using:

- Accuracy Score
- Confusion Matrix

Table 3: Model Comparison

Model	Accuracy	Remarks
Logistic Regression	(78.86%)	Baseline performance
Decision Tree	(77.24%)	Captures non-linearity
Random Forest	(78.05%)	Best performance

9. Results and Discussion

- Credit History dominates loan approval decisions
- Income variables require transformation due to skewness
- Decision Tree improves over linear models
- Random Forest provides the most reliable predictions

10. Conclusion

This project demonstrated how machine learning can be used to predict loan approval decisions effectively.

Exploratory Data Analysis revealed that **Credit History is the most influential factor**, followed by income-related variables and property area.

Feature engineering and log transformation improved data quality by reducing skewness and handling outliers.

Among the models tested, **Random Forest achieved the best performance**, highlighting the effectiveness of ensemble learning methods.

Overall, the study shows that data driven approaches can significantly enhance the efficiency and consistency of loan approval processes in financial institutions.